

# Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Micro-Project

PHASE 4 | PROJECT PLANNING PHASE

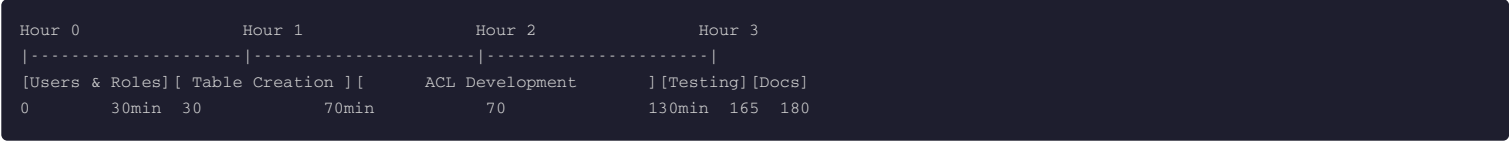
## 4. Project Planning Phase

### 4.1 Project Timeline

The micro-project is scoped to a total duration of **3 hours**, split into five sequential work blocks that mirror the milestone structure defined in the lab.

| # | Task Block                  | Key Activities                                                                        | Duration |
|---|-----------------------------|---------------------------------------------------------------------------------------|----------|
| 1 | Milestone 1 — Users & Roles | Create EEE User; create roles bb1-bb4; assign roles to user                           | 30 min   |
| 2 | Table Creation              | Create u_institution_details table; add 7 fields; create seed records for ECE/EEE/CSE | 40 min   |
| 3 | ACL Development             | Elevate security_admin; create Read, Create, Write, Delete ACLs with script logic     | 60 min   |
| 4 | Testing & Verification      | Impersonate EEE User and Administrator; verify each CRUD operation and branch filter  | 35 min   |
| 5 | Documentation & Wrap-up     | Capture screenshots, record outcomes, finalize project report                         | 15 min   |

### 4.2 Milestone Chart



### 4.3 Resource Plan

| Resource                      | Role in Project                                                       |
|-------------------------------|-----------------------------------------------------------------------|
| ServiceNow Developer Instance | Hosting environment for all configuration and testing                 |
| Admin account                 | Used to create users, roles, tables, and (after elevation) ACLs       |
| EEE User account              | Used to validate branch-restricted access through impersonation       |
| Documentation tooling         | Screenshot capture and report compilation across the 8 project phases |

### 4.4 Dependencies

- Role creation (bb1-bb4) must precede ACL creation, since each ACL references a role in its "Requires role" related list.
- Table and field creation must precede seed-record creation and ACL creation, since ACLs are defined against the table name.
- The security\_admin role must be elevated before any ACL record can be created or edited.
- Testing/verification depends on all four ACLs and all seed records (covering ECE, EEE, CSE) being in place.

### 4.5 Risk Register

| Risk                                                       | Likelihood | Impact | Mitigation                                                                                   |
|------------------------------------------------------------|------------|--------|----------------------------------------------------------------------------------------------|
| Forgetting to elevate security_admin before editing ACLs   | Medium     | Medium | Checklist step explicitly included before ACL creation (Milestone: Elevate role)             |
| ACL script blocks administrator access accidentally        | Low        | High   | Script always checks gs.hasRole('admin') first and returns true immediately                  |
| Missing seed data for one branch skews verification        | Medium     | Low    | Explicit step to create records for ECE, EEE, and CSE before testing                         |
| Role not assigned to test user                             | Low        | Medium | Role assignment step included immediately after role creation                                |
| Data Condition and Script conflicting (double restriction) | Low        | Medium | Design phase (3.4) documents the exact interaction between condition and script before build |

### 4.6 Success Criteria

- EEE User with bb1 role can view **only** the two EEE-branch records.
- A user without any bb role sees **zero** records ("No records to display").
- Administrator sees **all** records regardless of branch.
- EEE User with bb1+bb2 can view EEE records and use the **New** button to create records.
- EEE User with bb1+bb2+bb3 can additionally edit (write to) EEE records.
- EEE User with bb1+bb2+bb3+bb4 can additionally delete EEE records.

The plan sequences work so that identity and data foundations are always in place before security rules are layered on top, and reserves a dedicated testing block to validate every CRUD permission incrementally — directly supporting the 3-hour timer set for this micro-project.